

DSCI445 - Homework 2

Your Name

Be sure to `set.seed(400)` at the beginning of your homework.

```
#reproducibility
set.seed(400)
```

```
## load the data
library(MASS)
```

```
## take a look
head(Boston)
```

```
##      crim zn indus chas   nox    rm  age    dis rad tax ptratio  black lstat
## 1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3 396.90  4.98
## 2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8 396.90  9.14
## 3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8 392.83  4.03
## 4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7 394.63  2.94
## 5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7 396.90  5.33
## 6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7 394.12  5.21
##   medv
## 1 24.0
## 2 21.6
## 3 34.7
## 4 33.4
## 5 36.2
## 6 28.7
```

1. The MASS package contains a data set called `Boston` which records median house value (`medv`) for 506 neighborhoods around Boston. We will seek to predict median house value using 13 predictors, such as average rooms per house (`rm`), average age of houses (`age`), and percent of households with low socioeconomic status (`lstat`).

- (a) Start by visually inspecting the data to get an idea of relationships that might be present (**hint:** look into the `ggpairs` function in the `GGally` package.). Describe what you see.

```
## from the hint
library(GGally)
```

```
## Loading required package: ggplot2
```

```
## make plots and describe
```

- (b) For each predictor fit a simple linear regression model to predict the response. Describe your results. In which of the models is there a statistically significant association between the predictor and the response?

(b=c) Fit a multiple regression model to predict the response using all of the predictors. Describe your results (including diagnostic plots). For which predictors can we reject the null hypothesis $H_0 : \beta_j = 0$?

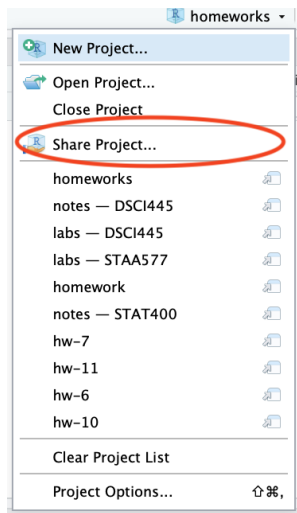
- (d) How do your results from (b) compare to your results from (c)? Create a plot displaying the

univariate regression coefficients from (b) on the x -axis and the multiple regression coefficients from (c) on the y -axis. That is, each predictor is displayed as a single point on the plot. Its coefficient in a simple linear regression model is shown as its x coordinate and its coefficient in a multiple linear regression model is shown as its y coordinate. Describe what you see.

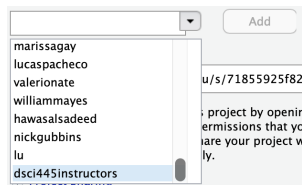
Turn in in a pdf of your analysis to canvas using the provided Rmd file as a template. Your Rmd file on the server will also be used in grading, so be sure they are identical.

Be sure to share your server project with the instructor and grader. You only need to do this once per semester.

1. Open your **homeworks** project on liberator.stat.colostate.edu
2. Click the drop down on the project (top right side) > Share Project...



3. Click the drop down and add “dsci445instructors” to your project.



This is how you **receive points** for reproducibility on your homework!